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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Academic Analytics for Institutional Decision Making

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

Educational institutions generate large volumes of student academic data every semester, including marks, attendance, grades, and pass/fail records. However, in many institutions, this data is primarily used only for result preparation and record keeping, rather than for performance analysis and decision-making.

One of the major challenges faced by academic institutions is the late identification of students who are at academic risk. By the time final results are announced, it becomes difficult to take corrective action for students with poor performance or low attendance. Institutions require a system that can continuously monitor academic performance and provide early insights.

In addition, academic management needs department-wise performance analysis, pass percentage reports, grade distribution analysis, and attendance impact analysis to support decisions such as remedial classes, attendance warnings, scholarship eligibility, and academic planning.

Currently, many colleges rely on manual Excel sheets and basic reports, which do not provide advanced insights or performance monitoring dashboards. Therefore, there is a need for an Academic Analytics System that can convert raw student data into meaningful insights and support institutional decision-making.

This project aims to develop an **Academic Analytics Dashboard using Microsoft Excel and AI tools** to analyze student performance, identify at-risk students, and support data-driven academic decisions.

- In colleges and universities, student academic performance data such as marks, attendance, and grades is collected every semester. However, this data is often stored in scattered Excel files or paper-based records, making it difficult for faculty and management to identify struggling students, compare departmental performance, or take timely corrective action.
- Academic institutions need reliable performance reports to support decisions related to re-examination policies, scholarship allocation, attendance warnings, and departmental improvements. Without structured data analysis, these decisions often depend on manual counting or guesswork, which is both time-consuming and error-prone.
- There is a clear gap in the use of data analysis tools at the college level. Most institutions have the data but lack the structured Excel dashboards,

pivot-based reports, and AI-assisted summaries needed to turn raw marks and attendance records into actionable academic insights. This project fills that gap by building a complete end-to-end analysis system using Microsoft Excel and Microsoft Copilot.

4. Objectives

- To collect and explore the raw dataset, understand its structure, fields, and data quality issues.
- To clean and preprocess the data by removing duplicates, handling missing values, correcting formats, and standardizing fields.
- To apply domain-relevant analysis using Excel formulas (SUMIF, COUNTIF, AVERAGEIF, VLOOKUP) and Pivot Tables to calculate student KPIs, grade distributions, and branch-wise performance comparisons.
- To build an interactive dashboard, structured report, or AI-generated output that communicates insights clearly to stakeholders.
- To integrate AI tools (Microsoft Copilot or AI Chatbot) to automate, enhance, or accelerate the analysis and content generation process.
- To document the complete project process, upload all files to GitHub, and record a demonstration video for submission.
- To calculate and compare key performance indicators (KPIs) such as overall pass rate, department-wise average scores, subject-wise performance, and grade distribution across all 1,000 student records using Excel formulas (COUNTIF, AVERAGEIF, SUMIF) and Pivot Tables.
- To examine the relationship between student attendance and academic performance, identify at-risk students with low attendance or failing grades, and present these findings through an Attendance vs. Performance analysis chart within the Excel dashboard.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description

1	**Data Collection & Exploration**	Obtain the raw dataset (CSV / Excel) from Kaggle, Edunet, or any suitable source. Understand field names, data types, and row count. For AI prompt projects, define the use case and target audience instead.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle null or missing values, fix incorrect data types, and standardize date and text formats using Excel (Remove Duplicates, IF, IFERROR, Text-to-Columns) or Power Query. For prompt-based projects, draft and structure initial prompts using Role, Context, Task, Format, and Tone components.
3	**Analysis & AI Integration**	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP), or Power BI DAX measures to derive key insights. Where applicable, use Microsoft Copilot or an AI Chatbot to generate narratives, summaries, or content outputs from the data or prompts.
4	**Visualization / Output**	Build interactive dashboards with charts, KPI cards, and slicers in Power BI or Excel. For AI-assisted projects, compile generated content outputs such as reports, marketing copy, or summaries. Document all AI prompts used alongside their outputs.
5	**Documentation & Submission**	Write the project report, upload all files to a GitHub repository, and record a video demonstration. Submit the GitHub link, video link, and dashboard link (if applicable) for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

A suitable public dataset for student academic performance analysis was not available on Kaggle or similar repositories that matched the required structure (department-wise, subject-wise, with attendance). Therefore, the dataset for this

project was manually prepared based on a sample college student data format, simulating realistic academic records typically found in Indian undergraduate institutions.

Dataset Summary:

- Total Records: 1,000 student entries
- Total Columns: 15 fields per record
- File Format: Microsoft Excel (.xlsx)
- Source: Custom-prepared dataset based on sample college academic records (not sourced from Kaggle)

Dataset Columns / Fields:

- Student ID — Unique identifier for each student
- Student Name — Full name of the student
- Gender – Male of Female
- Department / Branch — CSE, ECE, MECH, or CIVIL
- Subject 1 Marks, Subject 2 Marks, Subject 3 Marks — Individual subject scores (out of 100)
- Total Marks — Sum of all subject marks
- Average Score — Mean marks across all subjects
- Attendance (%) — Percentage of classes attended by the student
- Grade — Final grade assigned (A, B, C, D, E, F) based on average score
- Pass / Fail Status — Indicates whether the student passed or failed the semester

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*

2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 4*
5	Report writing, GitHub upload, and video recording and submission	*Week 5*

7. Expected Outcomes / Conclusion

This section summarizes what the team expects to deliver by the end of the project. Outcomes should be realistic and directly tied to the objectives stated above.

- A clean, validated dataset with all preprocessing steps documented and a before/after comparison.
- An interactive dashboard or structured analytical report that clearly communicates key insights from the data.
- Demonstrated use of an AI tool (Microsoft Copilot or AI Chatbot) to generate insights, summaries, or content outputs, with prompts and results documented.
- A GitHub repository containing all project files, a completed fellowship project report, and a video demonstration of the working output.
- A multi-sheet Excel workbook including a Raw Data sheet, Cleaned Data sheet, Pivot Tables sheet, Attendance Analysis sheet, and a fully formatted Main Dashboard sheet with KPI cards, bar charts, pie charts, and conditional formatting highlights.
- Key academic insights extracted from the data, including: overall pass rate, top-performing department, grade distribution across 1,000 students, subjects with highest and lowest average scores, and the impact of attendance levels on academic outcomes.
- An AI-assisted executive summary generated using Microsoft Copilot inside Excel, with documented prompts and responses demonstrating the practical use of generative AI in academic data reporting.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets
- Microsoft Corporation. Excel Functions Reference (COUNTIF, AVERAGEIF, VLOOKUP, SUMIF). Available at: https://support.microsoft.com/en-us/excel
- Chandoo.org. Excel Dashboard Tutorials and Pivot Table Guides. Available at: https://chandoo.org/wp/excel-dashboards/
- OpenAI. ChatGPT — AI Chatbot for Documentation and Content Generation. Available at: https://chat.openai.com
- Google. Google Gemini — AI Tool for Diagram and Report Generation. Available at: https://gemini.google.com